Zhenyao Cui

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Research Interests

Transfer learning; multimodal learning; neural decoding and BCIs; foundation models and AI agents.

Education

Huazhong University of Science and Technology Sep 2024 – Jun 2027
M.Eng. in Intelligent Science and Technology (expected)

- GPA **3.89/4.00**; selected for direct graduate admission through the national recommendation scheme.
- Advisor: Prof. Dongrui Wu, IEEE Fellow and Editor-in-Chief of *IEEE Transactions on Fuzzy Systems*.

Huazhong University of Science and Technology Sep 2020 – Jun 2024
B.Eng. in Automation, Elite Engineer Program

- GPA **3.91/4.00**; Outstanding Graduate of HUST.

Publications

† equal contribution.

1. **Zhenyao Cui**, Siyang Li, Siyuan Kan, Yudi Wang, Fei Luo, and Dongrui Wu. "A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces." *IEEE Transactions on Emerging Topics in Computational Intelligence*, Early Access, 2026. doi:10.1109/TETCI.2026.3683713
2. Siyang Li†, Jiayi Ouyang†, **Zhenyao Cui**†, Ziwei Wang, Tianwang Jia, Feng Wan, and Dongrui Wu. "Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces." *IEEE Journal of Biomedical and Health Informatics*, 2026. arXiv:2601.07556
3. **Zhenyao Cui**, Siyuan Kan, Siyang Li, Ziwei Wang, and Dongrui Wu. "SCORE: Subject Coordinate Recovery for Label-Free Cross-Subject EEG-to-Image Retrieval." Preprint, 2026; manuscript under review. arXiv:2608.19134.
4. **Zhenyao Cui**, Siyuan Kan, Dingkun Liu, and Dongrui Wu. "Beyond Trial Averaging: Anchoring Neural and Visual Representations for Few-Repetition Brain-to-Image Retrieval." Preprint, 2026; manuscript under review. arXiv:2608.19128.
5. **Zhenyao Cui**, Jiahui Chai, Siyuan Kan, Yudi Wang, and Dongrui Wu. "MCL-Align: Marginal–Conditional–Label Alignment for Cross-Subject EEG Regression." Manuscript under review, 2026.
6. **Zhenyao Cui**†, Siyuan Kan†, Lingfei Deng, Yunong Wang, Xuhua Ma, Xudong Zheng, and Dongrui Wu. "Domain-Pretrained Log Representations for Cloud Failure Prediction." Manuscript under review, 2026.
7. Siyuan Kan, Huanyu Wu, **Zhenyao Cui**, Fan Huang, Xiaolong Xu, and Dongrui Wu. "CMCRD: Cross-Modal Contrastive Representation Distillation for Emotion Recognition." *IEEE Transactions on Affective Computing*, 17(2):1837–1846, 2026. doi:10.1109/TAFFC.2026.3664341
8. Dingkun Liu†, Yuheng Chen†, Zhu Chen†, **Zhenyao Cui**, Yaozhi Wen, Jiayu An, Jingwei Luo, and Dongrui Wu. "EEG-FM-Compass: Progress, Benchmarking, and Future Directions for EEG Foundation Models." *National Science Review*, nwag466, 2026. doi:10.1093/nsr/nwag466
9. **Zhenyao Cui** (second author). "Heterogeneous Supervised Domain Adaptation (HSDA) for Cross-Device EEG Classification." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, manuscript under review; revision in progress. Contribution: theoretical analysis.

Patents

Dongrui Wu and **Zhenyao Cui**. "A Fuzzy-Set-Based Method for Constructing Regression Models for Vital-Sign Signals." (working English translation). Chinese invention patent **CN120449118B**, granted September 29, 2026. Assignee: Huazhong University of Science and Technology.

Research Experience

Huazhong University of Science and Technology
Apr 2024 – Present

Visual EEG Experiments and Neural Representation Analysis

Hands-on experience centers on EEG acquisition and BCI control software. Design and run EEG paradigms for image perception and image imagination. Collect EEG with Neuroscan and Neuracle systems using wet and dry electrodes, including a collaborator-developed local high-density array. Built a real-time SSVEP robotic-arm BCI with position-aware adaptive step sizing. Worked on continuous movement decoding from porcine ECoG.

- Co-authored **EEG-FM-Compass**; contributed to literature review and dataset curation, and benchmarked **LaBraM** and **CBraMod** for continuous PERCLOS prediction under cross-subject evaluation and limited calibration, comparing full fine-tuning with linear probing. *(National Science Review, 2026)*

Transfer Learning and Efficient Model Adaptation

Develop continuous-target transfer methods and efficient adaptation under changing users.

- Proposed **C2R**, using fuzzy sets to extend class-conditional distribution alignment from classification to regression. Enabled methods based on this alignment to handle continuous outcomes, and derived a target-domain risk bound to explain factors that support or limit transfer. *(first author, IEEE TETCI, Early Access, 2026)*
- Developed the regression method and experiments and contributed theoretical derivations for **BFT**, a framework that adapts predictions using transformed inputs without test-time backpropagation. *(equal-first author, IEEE JBHI, 2026)*
- Proposed **MCL-Align** for cross-subject EEG regression, jointly addressing marginal, conditional and label distribution differences. *(first author, under review)*

Visual Concepts and Cross-Modal Representation Learning

Study cross-person neural-visual representations and diagnose sources of retrieval errors.

- Proposed **SCORE** for label-free cross-subject EEG-to-image retrieval. Identified similar visual-concept relationships across users despite different representation coordinates. Used reliable neural–visual landmarks to recover target coordinates with frozen encoders, unlabeled target EEG and an image gallery. *(first author, under review)*
- Proposed **NEAR** for few-repetition brain-to-image retrieval. Target-swap experiments showed that query noise alone could not explain retrieval errors. Combined query denoising with image pseudo-anchor prediction guided by stable neural centers in EEG, and evaluated gallery-side correction in MEG and fMRI. *(first author, under review)*

Industry Experience

Alibaba Cloud Intelligence Group
Apr 2026 – Present

Research Intern

- **DPLR.** Co-developed representations pretrained on **167 million logs** with a domain tokenizer and masked language modeling, followed by time-aware attention and staged loss for prediction. The full system achieved **53.06% Max-F1**, **9.19 percentage points** above the internal baseline. A one-month production gray release identified about **125 additional servers that subsequently failed** with about **350 fewer false alerts**. *(equal-first author, under review)*
- **AutoResearch – Automated ML Experimentation.** Contributed to Alibaba Cloud's internal platform for agent-driven code modification and experiment evaluation. Tasks, code branches and metrics guide the process, with results informing subsequent attempts.

Dongfeng Motor Group – Multimodal Cockpit Evaluation
Jun 2025 – Jun 2026

Industry-funded project – Project Lead

- Led 32-channel dry-electrode EEG collection synchronized with video, audio and text to assess users' experience of in-vehicle functions.

Selected Projects and Honors

- **National Scholarship of China, 2026**; ranked first among the nominees in my major.
- **National Champion**, Brain-Controlled Robotic Arm track, China BCI Competition, 2025.
- **First Prize**, minimally invasive BCI track, World Robot Contest Championship, 2024. Developed porcine ECoG joint-motion decoding with segment-slicing augmentation.
- **First Prize**, National Undergraduate Intelligent Vehicle Competition, Multi-Vehicle Formation track, 2022.
- First-Class Graduate Academic Scholarship, HUST, 2024–25 and 2025–26; Yunnan Scholarship, 2024–25; Huawei Scholarship, 2021–22; Outstanding Graduate of HUST, 2024.